Arjun D. Patel

arjunpatel1215@gmail.com | 978-208-9749 | Greater Boston Area, Open to relocation

Experienced bioinformatics scientist skilled in NGS data analysis, pipeline optimization, and genomic insights. Expertise in Illumina, PacBio, and viral genomics to support advanced research and data-driven solutions.

EDUCATION

BOSTON UNIVERSITY

MS in Electrical and Computer Engineering
Dec 2018 | Boston, MA

BS in Biomedical Engineering
May 2017 | Boston, MA

LINKS

LinkedIn: /in/arjundpatel

SKILLS

PROGRAMMING

Python • R • Matlab •Tableau • HPC • Linux/Shell

NGS ANALYSIS

DNA • RNA • Viral • Illumina • PacBio • Variant Calling • NGS Visualization • Genome Databases • Bioinformatic Tools • NGS Pipeline

CORE

Data Management • Cloud Computing • Version Control • Collaboration • Communication

LANGUAGES

Gujarati

EXPERIENCE

MOLECULAR LOOP BIOSCIENCES | Bioinformatics Analyst
March 2022 – Present | Woburn, MA

- Processed and analyzed NGS data from Illumina and PacBio platforms for targeted sequencing across human and non-human genomes, delivering high-quality insights for clinical and research applications.
- Designed, implemented, and optimized bioinformatics pipelines for viral genomics projects, including SARS-CoV-2, influenza, and RSV, enabling scalable and reproducible analyses.
- Collaborated across multidisciplinary teams to develop customized workflows that addressed diverse sequencing objectives and project requirements.
- Performed comprehensive quality control, data normalization, and preprocessing to ensure integrity and accuracy in sequencing datasets.
- Conducted variant calling and CNV analysis, leveraging advanced algorithms and tools to extract actionable biological insights.
- Utilized IGV and other visualization tools to interpret NGS data, assess structural variants, and validate sequencing results.
- Created dynamic data visualizations to effectively communicate genomic findings and support scientific decision-making.
- Documented and maintained bioinformatics workflows to ensure reproducibility, scalability, and alignment with industry standards
- Analyzed viral genomic data to support public health initiatives, providing critical insights to support decision making and research efforts.
- Delivered results through detailed reports and presentations, effectively communicating findings to both internal and external stakeholders.
- Diagnosed workflow inefficiencies and implemented solutions to enhance data processing pipelines and streamline operations.
- Provided subject matter expertise in NGS data analysis and workflow development, aligning bioinformatics strategies with organizational objectives.

MERCK - MERCK RESEARCH LABORATORIES | Bioinformatics Intern
May 2018 – Aug 2018 | Boston, MA

- Analyzed and integrated genome wide association study summary (GWAS) statistics and expression quantitative trait loci (eQTLs) data to evaluate drug targets for neurodegenerative diseases using bioinformatics techniques.
- Created a bioinformatic pipeline

PHYSICAL SCIENCES INC. | Intern
July 2016 – Aug 2016 | Andover, MA

- Defined and improved a scoring algorithm to differentiate between tumor, adipose and fibrous tissue, and cancerous and noncancerous using MATLAB.
- Used optical coherence tomography (OCT) and low-coherence interferometry (LCI) techniques to successfully aid tissue discrimination.

PHYSICAL SCIENCES INC. | Intern
May 2015 – Aug 2015 | Andover, MA

- Programmed for ophthalmic imaging technologies using MATLAB.
- Used high-speed image processing along with many other image processing techniques to turn medical images (optical coherence tomographs, ultrahigh resolution photographs of cone photoreceptors, etc.) into research data.